PCI AI · FORMULA SHEET

The Project Finance Formula Sheet

Every formula that decides whether a project is bankable

NPV · IRR · WACC · CFADS · DSCR · LLCR · PLCR

PROJECT CONTROLS INSTITUTE GLOBAL

First edition · 2026 · AI proposes. The professional disposes.

What this is

Forty-one formulas across project and infrastructure finance: time value, appraisal, cost of capital, the cash waterfall, coverage ratios, debt sizing and sculpting, equity returns and construction drawdown.

Every one is drawn from the PFL-AI Body of Knowledge and carries the domain where it is worked in full.

Against each, the error it attracts — because in project finance most wrong answers come from a definition or a period convention, not from the expression.

Save this one. It is the reference, not a highlight reel.

CONVENTIONS

The rule that catches most model errors

Rates, periods and compounding must agree.

An annual rate applied to semi-annual debt service is the single error the verification suite catches most often across the whole programme.

And arithmetic is decimal, never binary floating point. Intermediate values carry full precision; rounding happens at display only — which is why a printed figure will not reproduce if you round as you go.

01

Time value of money

Domain 3. The machinery every other number in this deck is built on.

DOMAIN 3 · CORE

Discounting and compounding

EXHIBIT 1

FORMULA	MEANING	DOMAIN
FV(x) = PV(x) × (1 + r)^n	Compounding	3.1
PV(x) = FV(x) ÷ (1 + r)^n	Discounting	3.1
DF(t) = 1 ÷ (1 + r)^t	Discount factor at period **t**	3.1
AF(r, n) = [1 − (1 + r)^-n] ÷ r	Annuity factor	3.2
A = Principal ÷ AF(r, n)	Annuity payment	3.2

A free check. The annuity factor must equal the sum of the individual discount factors over the same periods. If the two disagree, your period convention is wrong before anything else is.

DOMAIN 3 · EXTENSIONS

Perpetuities and real rates

EXHIBIT 2

FORMULA	MEANING	DOMAIN
PV(perpetuity) = CF ÷ r	Level perpetuity	3.2
PV(growing perpetuity) = CF_1 ÷ (r − g)	Gordon growth; valid only while **g < r**	3.2
1 + i_nom = (1 + i_real) × (1 + π)	Fisher relation	3.4
Effective annual rate = (1 + r/m)^m − 1	**m** compounds per year	3.3

Fisher is multiplicative, not additive. Subtracting inflation from the nominal rate is close enough at 2% and materially wrong at 20% — which is precisely where the projects that need it are.

02

Investment appraisal

Domain 4. Whether the asset is worth building at all.

DOMAIN 4 · APPRAISAL

The decision measures

EXHIBIT 3

FORMULA	MEANING	DOMAIN
$NPV = \Sigma CF_t \div (1 + r)^t - I_0$	Net present value	4.1
IRR: NPV = 0	Internal rate of return	4.1
MIRR	Modified IRR — explicit reinvestment rate	4.1
$PI = PV(\text{future CF}) \div I_0$	Profitability index	4.2
$EAV = NPV \div AF(r, n)$	Equivalent annual value — unequal lives	4.3
Payback	Period to recover the outlay	4.2

PI also equals $1 + NPV \div I_0$. Compute it both ways as a check; if the two disagree, a cash-flow sign is wrong.

DOMAIN 4 · IRR PATHOLOGIES

Three ways IRR misleads

Multiple roots. More than one sign change in the cash flows means more than one IRR. Check the sign pattern before quoting one.

No solution. If the flows never cross zero there is nothing to solve for.

Scale and reinvestment blindness. IRR implicitly assumes reinvestment at itself — which is exactly why **MIRR** exists.

> **On mutually exclusive projects, NPV governs.** The higher IRR is regularly the smaller prize, and IRR cannot see that.

03

Cost of capital

Domain 9. The rate everything above is discounted at.

DOMAIN 9 · WACC AND GEARING

Capital structure

EXHIBIT 4

FORMULA	MEANING	DOMAIN
$WACC = g \times k_d \times (1 - T) + (1 - g) \times k_e$	Cost of capital, in gearing form	9.1
$g = D \div (D + E)$	Gearing	9.2
D/E	Debt to equity	9.2
$k_e = R_f + \beta \times (R_m - R_f)$	CAPM cost of equity	9.1
$\beta_{asset} = \beta_{equity} \div [1 + (1 - T) \times D/E]$	Ungearing beta	9.1

Two cautions. **(1 − T)** attaches to the cost of **debt** only — the tax shield never touches the equity term. And WACC is linear in gearing only while **k_e** is held fixed; in reality **k_e** rises with leverage, which is what the ungearing formula exists to handle.

04

Cash flow and the waterfall

Domains 2, 6 and 14. What the ratios are actually measured on.

DOMAINS 6, 14 · WATERFALL

The order the money moves

EXHIBIT 5

	PAID IN THIS ORDER
1	Operating expenses
2	Tax
3	Senior debt interest
4	Senior debt scheduled principal
5	Reserve top-ups — DSRA, MRA
6	Subordinated debt service
7	Distributions to equity, subject to lock-up

Each level is paid only after the one above it is satisfied.

DOMAIN 2 · CFADS

The most consequential definition in the model

CFADS = EBITDA − Tax paid − Δ Working capital − Maintenance capex

Note what is **not** deducted: interest. Interest and principal are what CFADS is measured against.

CFADS is a defined term, not a standard one. Whether it is struck before or after working-capital movements changes every ratio built on it — so the definition belongs in the term sheet, agreed, not assumed. Deduct interest above the line and you double-count it, inflating every cover ratio in the direction that makes a deal look financeable.

DOMAIN 10 · STRUCTURE

Reserves and triggers

EXHIBIT 6

FORMULA	MEANING	DOMAIN
DS = Interest + Scheduled principal	Debt service	10.2
DSRA = Debt service × months ÷ 12	Reserve, expressed as the shortfall it survives	10.3
Lock-up trigger in cash = Debt service × threshold ratio	The number that belongs on a dashboard	10.3
Cash to equity = CFADS − DS − Reserve top-ups	Distributable cash	6.4

Put the covenant on the dashboard in cash, not as a ratio. "DSCR must exceed 1.20" is an abstraction. "We must generate 33m this period" is a number an operations team can actually act on.

05

Coverage ratios

Domain 10 — the quantitative flagship. The three questions every lender asks.

DOMAIN 10 · THE THREE RATIOS

Cover, in one period and over a lifetime

EXHIBIT 7

FORMULA	MEANING	DOMAIN
DSCR = CFADS ÷ (Interest + Scheduled principal)	Cover in a single period	10.2
LLCR = PV(CFADS over loan life, at the loan rate) ÷ Outstanding debt	Cover over the life of the loan	10.2
PLCR = PV(CFADS over project life) ÷ Outstanding debt	Cover over the life of the project	10.2
ICR = EBIT (or EBITDA) ÷ Interest	Accounting cover; ignores principal entirely	10.2

The minimum DSCR is the covenant. The average never is. A comfortable average conceals the single period that breaches — and the single period is what triggers lock-up.

THE TAIL

Why all three exist

Tail = Project life − Loan life

PLCR exceeds **LLCR** wherever a tail exists, because it discounts a longer stream of cash against the same debt balance.

The gap between the two ratios is the tail — the project's cash-generating life after the debt matures, and therefore the lender's room to reschedule rather than enforce. A thin tail is a credit concern even when LLCR looks comfortable.

DOMAIN 10 · DEBT CAPACITY

Sizing the debt backwards

EXHIBIT 8

FORMULA	MEANING	DOMAIN
Max debt service = CFADS ÷ Target DSCR	Affordable service per period	10.1
Max debt capacity = Max debt service × AF(r, n)	Quantum at the loan rate and tenor	10.1
Sculpted debt service_t = CFADS_t ÷ Target DSCR	Coverage constant by construction	10.1
Average debt life = Σ (Principal_t × t) ÷ Σ Principal_t	Weighted average life	10.1

Check it by reversing. Service exactly the sized capacity and the DSCR must land on the target in every period. If it does not, the capacity and the discounting are on different period conventions.

REPAYMENT SHAPE

Sculpted or annuity

Sculpted sets debt service as a constant multiple of CFADS, so DSCR is flat at the target in every period.

Annuity sets a constant instalment, so DSCR rides up and down as CFADS moves.

> **Sculpting maximises debt capacity for a given minimum DSCR.** That is why limited-recourse project finance uses it and corporate lending largely does not — and why, on a sculpted profile, the minimum DSCR is the only ratio worth arguing about.

06

Returns and construction

Domains 14 and 15. What equity earns, and what the build actually costs to fund.

DOMAINS 6, 15 · RETURNS

Two IRRs, two questions

EXHIBIT 9

FORMULA	MEANING	DOMAIN
Project IRR	Return on total project cost, pre-financing	4.1
Equity IRR	Return on equity cash flows only	6.4
MOIC = Total distributions ÷ Total equity invested	Money multiple; ignores timing entirely	15.2

Project IRR asks whether the asset is worth building. Equity IRR asks whether the deal is worth doing. A strong equity IRR on a weak project IRR is a statement about leverage, not about the asset — so quote both, or you have quoted neither.

DOMAIN 14 · CONSTRUCTION

What the build really costs to fund

EXHIBIT 10

FORMULA	MEANING	DOMAIN
IDC = Σ Interest on the drawn balance during construction	Interest during construction	14.2
Total project cost = Construction + IDC + Fees + Working capital + Reserves	The funding requirement	14.2
Drawdown_t = Cost incurred_t × Debt proportion	Pro-rata draw	14.1
Cost to complete = Total budget − Costs incurred	Remaining requirement	14.3
Funding adequacy: Available funds ≥ Cost to complete	The lender's construction test	14.3

The bridge to project controls. A lender's cost-to-complete test and a controls team's **EAC** answer the same question from two sides. If controls forecasts an outturn above the funded amount, funding adequacy has already failed — usually before anyone in the finance team has been told.

DOMAIN 11 · RISK

Sensitivity and the downside case

EXHIBIT 11

FORMULA	MEANING	DOMAIN
EMV = Probability × Impact	Expected monetary value	11.2
Switching value = Δ% that drives NPV or minimum DSCR to its limit	Breakeven sensitivity	11.3
Breakeven tariff / volume	Price or throughput at the covenant	7.2

The standard set: capex overrun, delay to commercial operation, revenue or tariff reduction, opex increase, interest-rate movement, availability shortfall. Each run singly, then combined into one downside case — because in reality they correlate and never arrive alone.

MOST MISAPPLIED · 1–5

The ten that cost the most

1. Interest deducted before **CFADS** — every ratio inflated
2. Annual rate applied to semi-annual debt service
3. **CFADS** struck on an undefined working-capital basis
4. **LLCR** and **PLCR** treated as interchangeable
5. Average DSCR quoted as the covenant

MOST MISAPPLIED · 6–10

And the other five

1. **IRR** ranked above **NPV** on mutually exclusive projects
2. Multiple IRRs unnoticed — more than one sign change, more than one root
3. Tax shield applied to the equity term of **WACC**
4. **IDC** and fees omitted from total project cost
5. Equity IRR presented as project performance

Each of these survives model review, because the spreadsheet computes cleanly and the output looks entirely plausible.

The Body of Knowledge is open

Every formula here is developed to full depth in the PFL-AI Body of Knowledge — sixteen domains with worked examples, master-model threads and calculation exercises, at the domain cited on each slide. Domain 10 carries coverage ratios, sculpting and covenants as the quantitative flagship.

Published openly. **No registration, no email gate.**

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